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## Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Brynge M, Sjöqvist H, Gardner RM, Lee BK, Dalman C, Karlsson H. Maternal infection during pregnancy and likelihood of autism and intellectual disability in children in Sweden: a negative control and sibling comparison cohort study. *Lancet Psychiatry* 2022; published online Sept 7. https://doi.org/10.1016/S2215-0366(22)00264-4.

**Supplementary Method**

We followed all children from birth to diagnosis of autism or ID, death, emigration out of Stockholm, their 18th birthday, or 31 December 2016, whichever came first. We used Cox regression models assuming a proportional hazard from 0-18 years (i.e., the weighted average of the risk during childhood), to model potential associations between infection and autism and/or ID.

The following two approaches were implementing to account for the potential unmeasured confounding:

1) Negative control exposures are used to detect bias from unmeasured confounding by comparing an exposure of interest with an alternative exposure (the negative control), with similar associations with the unmeasured confounders, but with no plausible causal pathway to the outcome.[2] If the association between the exposure of interest and the outcome is indeed caused by unmeasured confounding factors, then the negative control exposure will also be associated with the outcome, despite a lack of a causal pathway to the outcome, due to its similar relations with those confounding factors. As a negative control, we measured the risk of autism or ID associated with maternal infections during a 40-week period one year before pregnancy (i.e., the analogous time period and seasonality for the index pregnancy but occurring in the year prior to the index child's conception). This resulted in a three-month window between the last day of at-risk for infection in the negative control analysis and start of the index pregnancy and time at risk in the main analysis. Models were adjusted similarly to the pregnancy models, including adjustment for paternal infection during the same period and for maternal infection during pregnancy to account for differences in rates of infections among parents to individuals with autism and/or ID.

We also considered using the paternal infection as a negative control but abandoned the idea for a likely and potential cross-infection between the parents – e.g., the mother is likely to infect the father and vice-versa.

2) We conducted a survival-based sibling analysis, where we compared the occurrence of maternal infection among sibling-pairs of the same biological mother. If estimates from a sibling comparison model are comparable to the population-based comparison, this provides evidence that the population-based estimates are not likely affected by unmeasured shared familial factors. We used a Cox regression model stratified on the biological mother. We compared the estimates of the stratified (within family) model to estimates from a parametric model (between family comparison, as described for the total population above) restricted to the sibling sample, to examine if selection of the sibling sample might be a cause of a potential bias towards the null. To examine a potential deviation in the risk over time (non-proportional), we also estimated the scaled Schoenfeld residuals. By modelling the association between the scaled residuals and time-to-event we were able to fit an approximate hazard ratio for the individual's follow-up during childhood that could vary over time, which was reported alongside the stratified proportional hazard ratio estimates. To test for carryover effects of maternal infection of the pregnancy of the first sibling to autism or ID in the second sibling,[3] we estimated associations between maternal infection and autism or ID in two separate sub-samples of the sibling cohort: a) Restricted to older exposed/younger unexposed, and b) Restricted to older unexposed/younger exposed. By dividing the sibling sample accordingly and comparing the estimates, we were also able to assess potential bias from misclassification of the outcomes due to diagnostic time-trends.

To examine a potential non-linear relationship over the age of the child, we implemented a flexible parametric survival model using restricted cubic splines to model non-linear associations for the baseline distribution function and a possible time-dependent effect of the exposure, with the number of knots for each analysis determined by number of events and pattern.[1]

**References**


1. Lambert PC, Royston P. Further development of flexible parametric models for survival analysis. *Stata Journal* 2009; **9**(2): 265-90.
2. Lipsitch M, Tchetgen Tchetgen E, Cohen T. Negative controls: a tool for detecting confounding and bias in observational studies. *Epidemiology* 2010; **21**(3): 383-8.
3. Sjölander A, Frisell T, Kuja-Halkola R, Öberg S, Zetterqvist J. Carryover Effects in Sibling Comparison Designs. *Epidemiology* 2016; **27**(6): 852-8.

**Figure S1**. The adjusted relationships between maternal infection during pregnancy and risk of ID and autism, over age at diagnosis. Models were adjusted for children's sex, birth year, and season of birth; maternal BMI, age at delivery, region of origin, psychiatric history, and pre-pregnancy infection; paternal region of origin, infection during the same time period, and psychiatric history; parental education, and income. For both ID and autism, the number of degrees of freedoms (default placements) at the baseline distribution were 3 and for the time-depending effect of infection we used 2 degrees of freedom. With the parametric model, we fitted an non-linear model over time to estimate the hazard ratio over the course of follow-up, represented by the solid line, and the 95% confidence interval of the hazard ratio, represented as the shaded bands around the solid line.

The red dots indicate when during follow up the specific diagnosis of the neurodevelopmental disorder occurred for children exposed (top) and unexposed (bottom) to maternal infections during pregnancy. The proportional hazard ratio indicates the hazard ratio under the proportional hazard assumption in a Cox regression model, or the weighted average of the HR over the course of follow up. Abbreviations: **BMI**: Body mass index; CI: Confidence interval; **HR**: Hazard Ratio; **ID**: Intellectual disability.

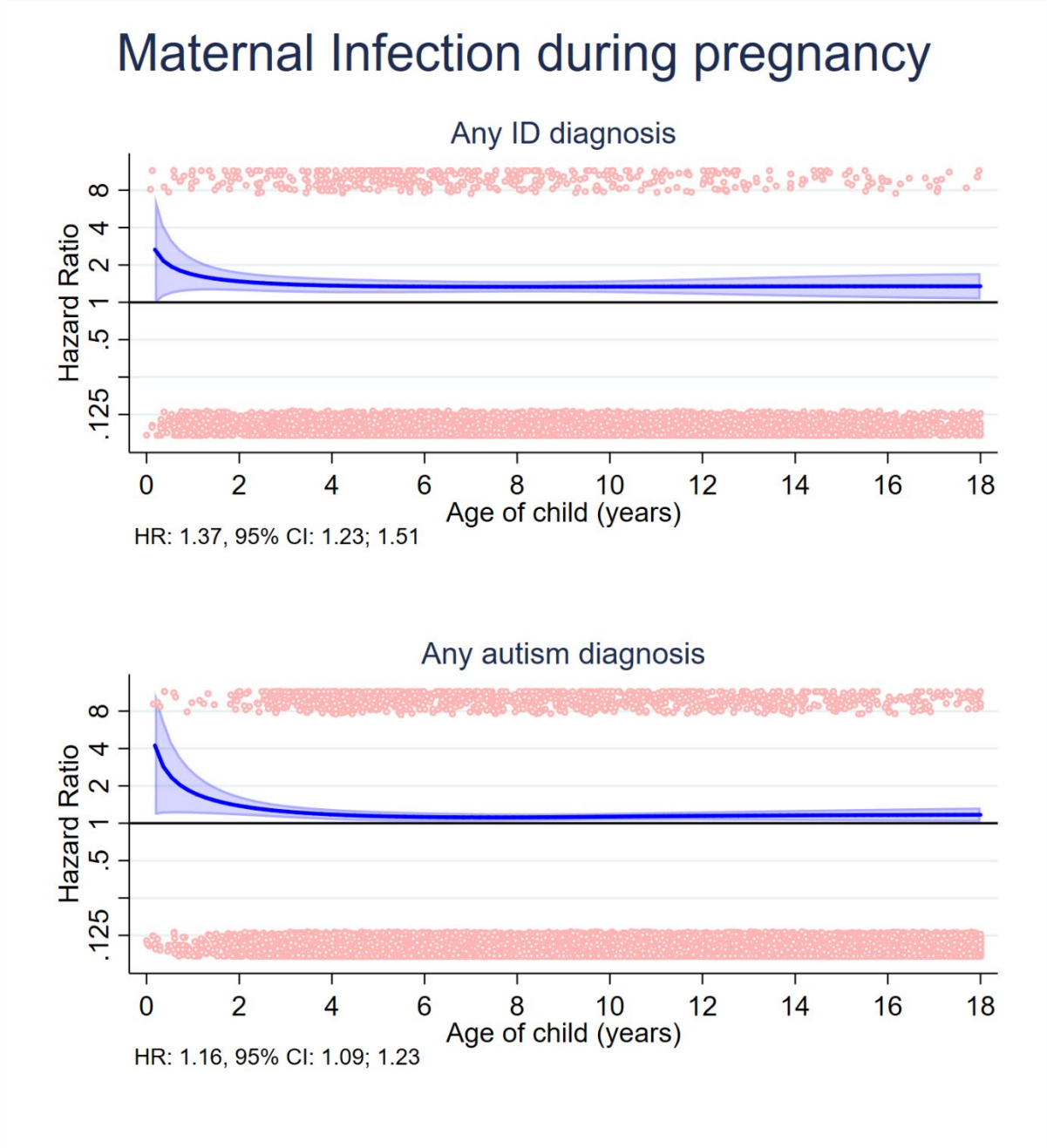
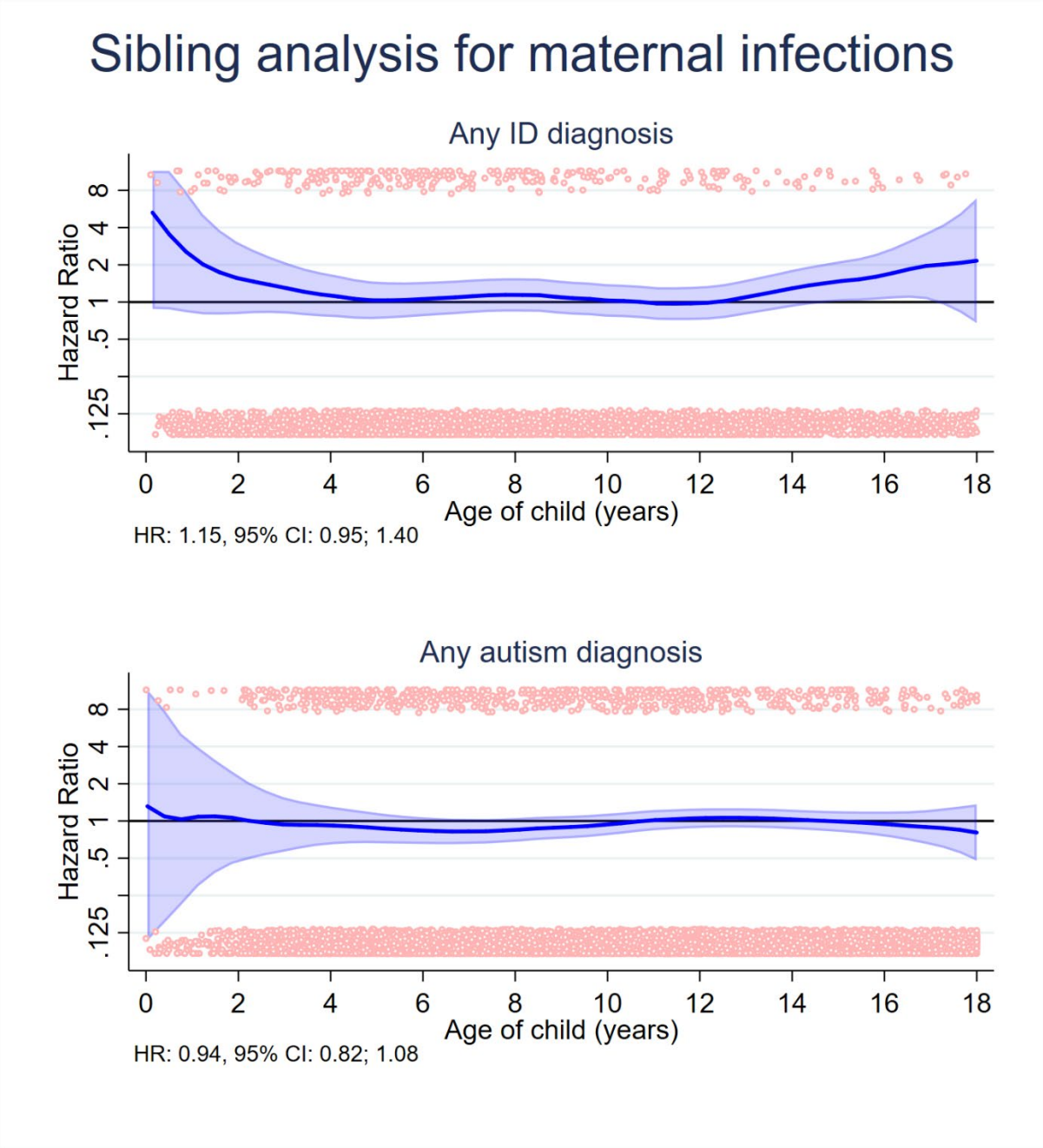
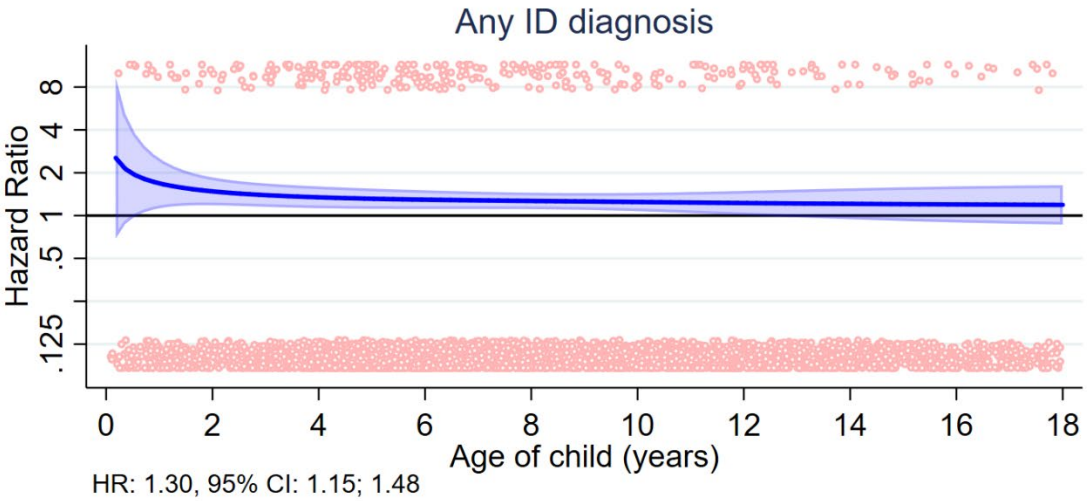


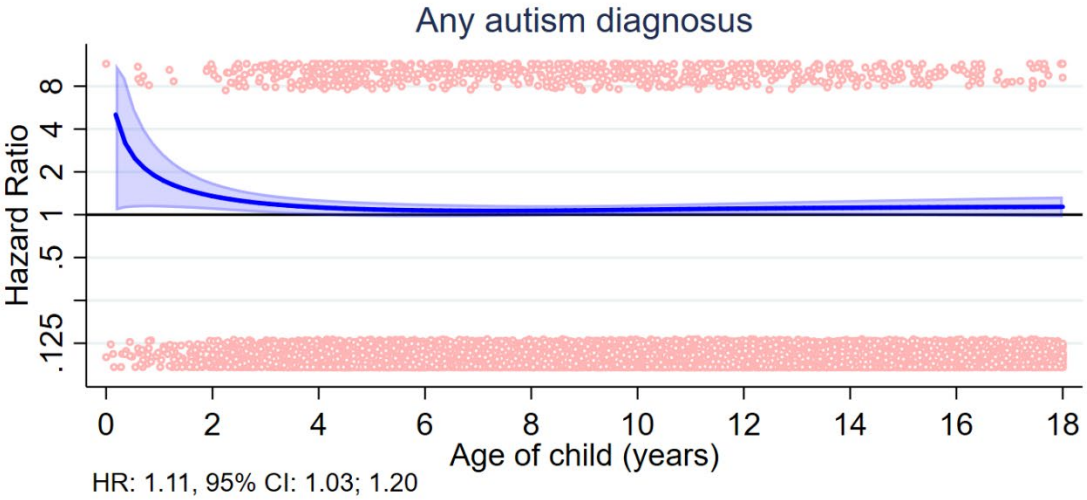
**Figure S2**. The adjusted relationships between maternal infection during pregnancy and risk of ID and autism, over age at diagnosis, when comparing individuals diagnosed with ID or autism with their unaffected siblings. Models were stratified on nuclear family and adjusted for children's sex, birth order, birthyear, and season of birth; and paternal infection. Using the scaled Schoenfeld residuals, we fitted an approximate non-linear model over time to estimate the hazard ratio over the course of follow-up, represented by the solid line, and the 95% confidence interval of the hazard ratio, represented as the shaded bands around the solid line. The red dots indicate when during follow up the specific diagnosis of the neurodevelopmental disorder occurred for children exposed (top) and unexposed (bottom) to maternal infections during pregnancy. The proportional hazard ratio indicates the hazard ratio under the proportional hazard assumption in a Cox regression model, or the weighted average of the HR over the course of follow up. Abbreviations: **CI**: Confidence interval; **HR**: Hazard Ratio; **ID**: Intellectual disability.



# Parametric Model on Sibling sample

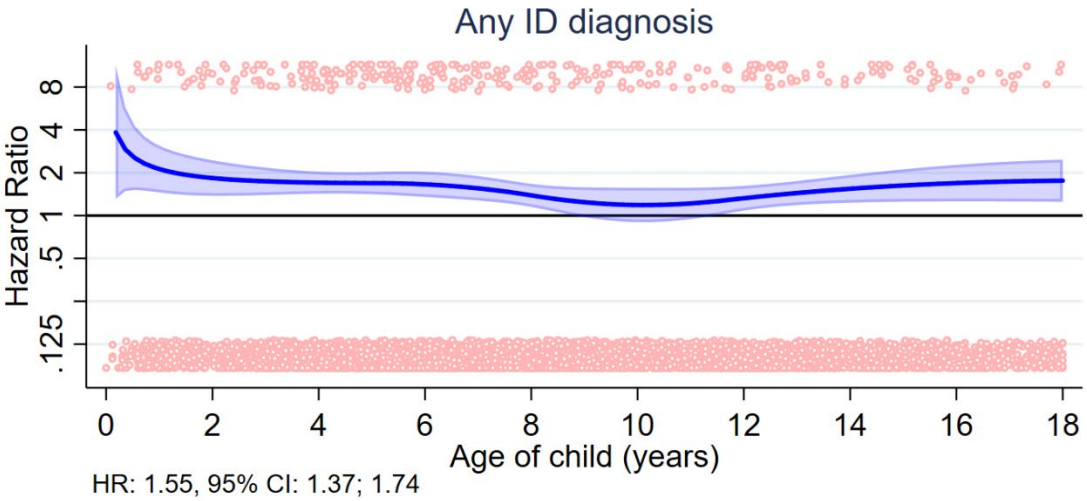



HR: 1.30, 95% CI: 1.15; 1.48

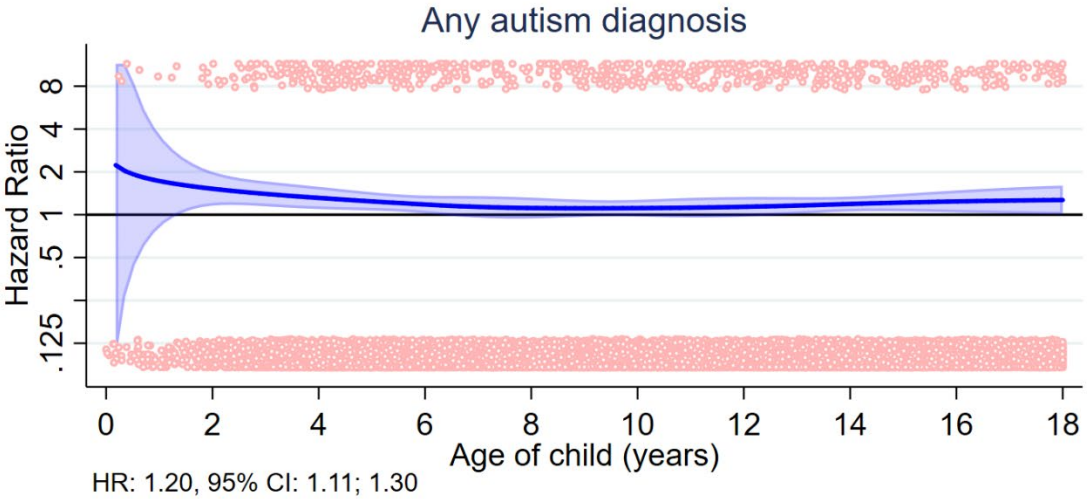



HR: 1.11, 95% CI: 1.03; 1.20

# Maternal Inpatient Infection

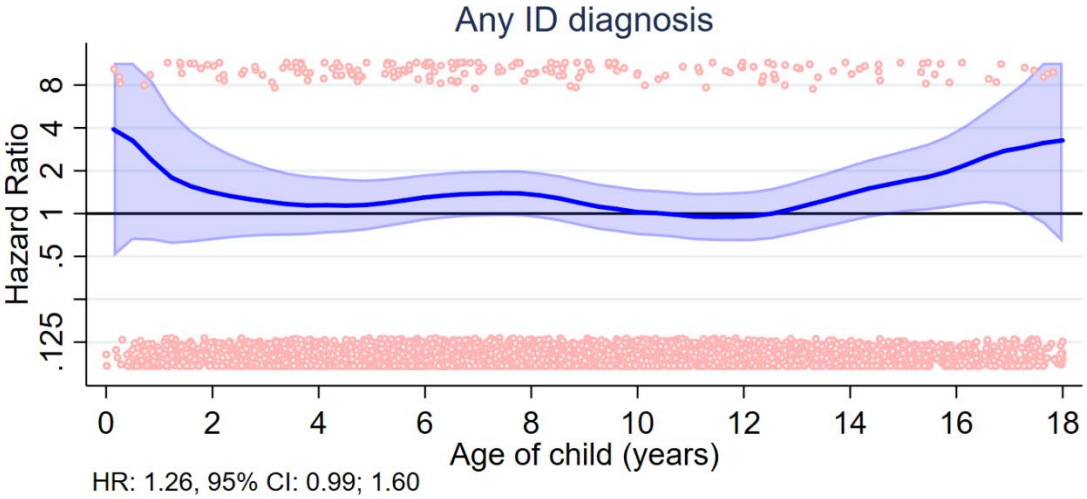



HR: 1.55, 95% CI: 1.37; 1.74

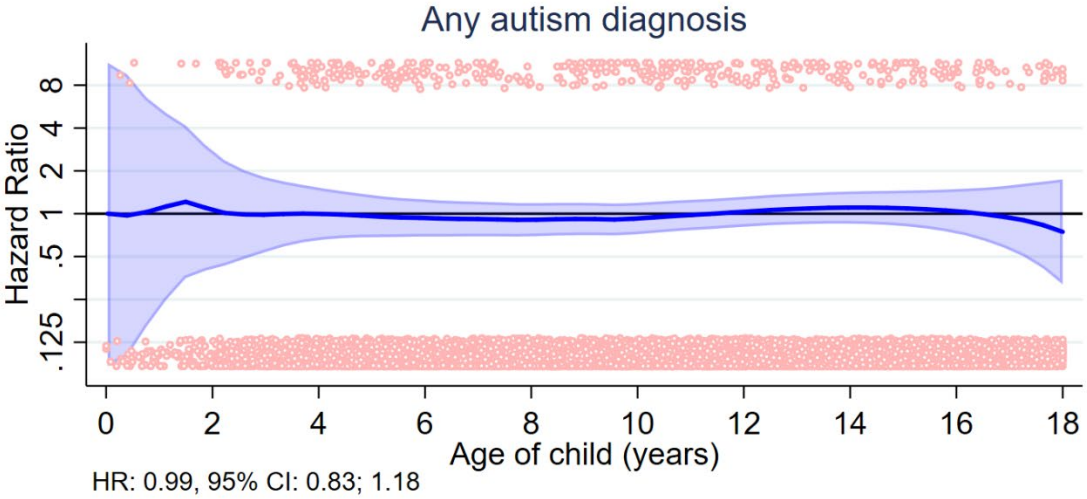



HR: 1.20, 95% CI: 1.11; 1.30

# Sibling analysis for Inpatient infections




HR: 1.26, 95% CI: 0.99; 1.60




HR: 0.99, 95% CI: 0.83; 1.18

**Figure S6**. Distribution of registered infections over birthyears and across the different register sources.

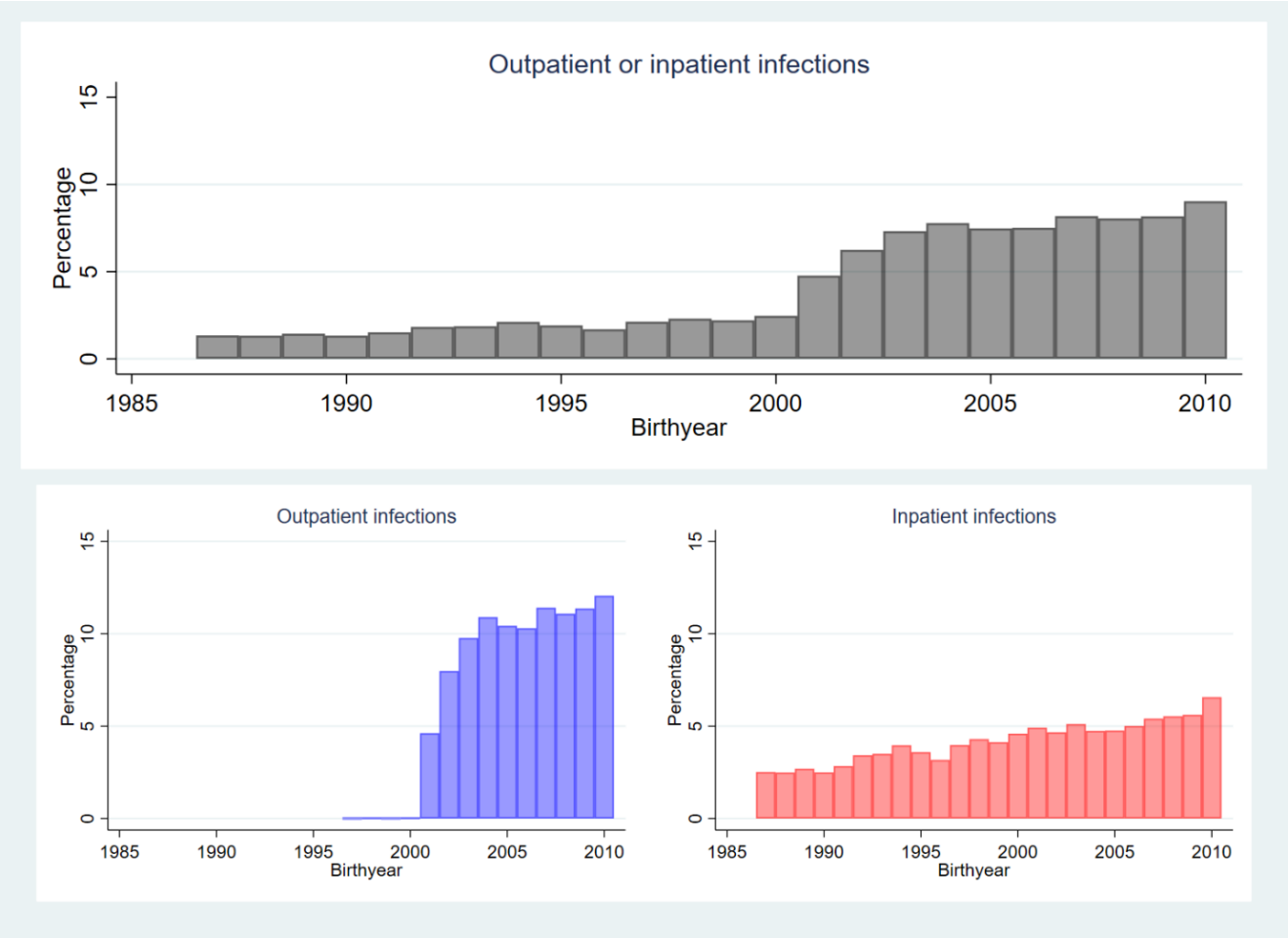


**Table S1.** Diagnostic codes and register databases used to ascertain autism, ID, and parental psychiatric diagnoses, within the Stockholm Youth Cohort (SYC).

| Registers | Covariates Derived | Diagnostic Codes |
|---|---|---|
| Registry of Total Population (1968-2016)<br><br>Multi-generation Registry (as of 2010) | • Birth year<br>• Identity of biological mother and father<br>• Sex<br>• Birthplace in Sweden<br>• Parental migration status<br>• Child's migration date (censoring) | |
| | | |
| Cause of Death Register (1961-2016) | • Date of death (censoring) | |
| | | |
| The Longitudinal Integrated Database for Labor Markets Studies (LISA; 1990-2016) | • Parental education<br>• Parental income | |
| | | |
| National and Regional Health Registers[a,b,c,d,e] | • Autism[a,b,c,d] | ICD-9: 299;<br>ICD-10: F84;<br>DSM-IV: 299; |
| | • Intellectual Disability[a,b,c,d] | ICD-9: 317-319;<br>ICD-10: F70-F79;<br>DSM-IV: 299; |
| | • Parental Psychiatric Diagnoses[a,c,e] | ICD-8: 290-315;<br>ICD-9: 290-315;<br>ICD-10: F-chapter diagnosis;<br>DSM-IV: 290-315 |
| | • Congenital/Metabolic Disorders[a] | ICD-9: 758, 759F, 237H, 270;<br>ICD-10: Q85.0, Q85.2, Q90-99, E70-72 |
| | | |
| Medical Birth Register (1973-2010) | • Maternal height<br>• Maternal weight (first antenatal visit)<br>• Birth order | |

**Note:**

a. The National Patient Register (NPR): including inpatient care from 1973, outpatient physician visits in specialist care from 1997.

b. Stockholm Clinical Database for Child and Adolescent Psychiatry: DSM-IV was used until 2008; ICD-10 was used thereafter.

c. VAL register: Stockholm county health care databases including in- and outpatient care, regardless of specialty/primary health care.

d. The Habilitation Register (HAB): Provides data on utilization of Stockholm County Habilitation services according to type of disability, such as pervasive developmental disorders and intellectual disability (recorded as present or absent). HAB register was used until 2008; since 2008 it has been included in the VAL databases as ICD-coded diagnoses (see C above).

e. Psykiatrivårdsystemet (PVS) Stockholm (Psychiatric Care System).

**Table S2**. ICD-codes for the classification of the infectious exposures.

| Type of Infection | ICD Version[a] | Diagnostic Codes for specialized care for infections |
|---|---|---|
| Any infection | ICD-8 | 006.00-007.99, 009.00-009.98, 084.00-087.99, 099.96-099.99, 110.00-130.10, 130.99-131.99, 136.09, 320.88-320.99, 360.00, 380.02-381.99, 384.00-384.08, 420.00-420.09, 421.98, 422.97-422.99, 462.01, 462.09, 463.09, 466.99, 483.99-486.09, 503.00-503.09, 540.00-540.02, 540.04-540.99, 572,99, 686.00-686.98, 761,40, 763.10, 763.98, 778,60 + ICD-8 codes in 'Bacterial infection' and 'Viral infection'. |
| | ICD-9 | 006-007X, 008W, 009-D, 084-086X, 099E-X, 110-136X, 321A, 321W, 370E-F, X, 372A-D, 380B, C, 381A, 382X, 420- 422X, 462-463, 466-B, 473-X, 483, 485-486, 490, 491B, 540A, X, 572A, 647C, E, W, X, 680A, 711G-X, 727A, 770A, 771C, E-W + ICD-9 codes in 'Bacterial infection' and 'Viral infection'. |
| | ICD-10 | A06-07.9, A08.5, A09, A59-59.9, A63, A63.8-64, B35-49, B50 -89, B99, G02.1-02.8, G04, G04.9, G05.2, H10.0, H10.3-10.9, H16.2-16.3, H16.9, H32, H60, H60.3, H65.0-65.1, H66.9, I30.0-30.9, I33.0-33.9, I40.0, J02[b], J02.8-02.9[b], J03[b], J03.8-03.9[b], J16, J16.8, J18-18.9, J20, J20.8-21, J21.8-21.9, J22, J32-32.9[b], J35.0, J37-37.1[b], J40-42, K35, K35.9, K75.0, L30.3, M46.5, M65.1, M71.1, O98.3, O98.6-98.9, P23.8-23.9, P37.1-39.9, Z22.4, Z22.8-22.9 + ICD-10 codes in 'Bacterial infection' and 'Viral infection'. |
| Bacterial | ICD-8 | 000.01-005.99, 008.00-008.30, 010.99-018.98, 020.00-039.98, 073.99, 076.99, 079.30, 080.99-083.99, 088.99-104.98, 320.00-320.80, 322.00-322.03, 361.00-361.09, 362.02, 366.00, 369.00, 380.00-380.01, 382.00-383.99, 390.97-392.99, 421.00, 461.00-461.09, 462.02, 463.01, 481,99-482.98, 501.99, 508.00-508.02, 510.01-510.09, 511.10, 513.99, 522.50, 527.30, 528.00, 528.30, 540.03, 562.00-562.19, 566.00-566.01, 567.00-567.02, 569.00, 577.01, 590.00-590.99, 595.00-595.02, 597.00, 599.02, 611.00, 611.01, 612.01-614.99, 616.00-616.03, 620.00-620.99, 622.00-622.19, 629.40, 630.00-630.09, 635.00-636.09, 645.90-645.91, 670.00-670.09, 678.02, 680.00-682.99, 684.00-684.09, 710.00-710.09, 720.00-720.29, 732.99, 761,00, 763.00, 998.50, 999.30 |
| | ICD-9 | 001-005X, 008A-F, 010-041X, 073, 076, 078D, J, 790H, 080-083X, 087-099D, 100-104, 245A, 254B, 320-X, 324-X, 360A, 373B, 375D, 376A, 382A-E, 383A-X, 390-392X, 421A, 461-X, 475, 481-482X, 510-X, 511B, 513-B, 522E, H, 526E, 527D, 528A, D, 540B, 562-B, 566, 567-C, 569F, 575A, 590-X, 597A, 595-D, X, 597W, 599A, 611A, 614-F, W-X, 615A, X, 616-X, 634A, 635A, 636A, 637A, 638A, 639A, 646F, G, 647A, B, D, 658E, 659D, 670, 675-B, W-X, 681-686X, 711A, E, 728A, 729E, 730-D, X, 771D, 996G, 998F, 999D |
| | ICD-10 | A00-05.9, A15-17.9, A20-28.9, A30 -58, A65 -79.9, B95-96.8, E06.0, E32.1, G00-00.9, G01, G04.2, G05.0, G06-06.2[b], G07, H00.0, H01.0, H04.3, H05.0, H44.0, H60.0-60.1, H66.0-66.4, H70.0-70.9, I00-02.9, J01-01.9[b], J02.0, J03.0, J13-15.9, J16.0, J20.0-20.2, J34.0, J36[b], J39.0-39.1, J85.1-85.3, J86-86.9[b], K04.6-04.7, K05.2, K11.3, K12.2, K14.0, K35.1, K57-57.9, K61-61.4, K63.0, K65.0[b], K81.0, K85, L00 -08.9, M00-00.9, M46.3[b], M60.0[b], M86-86.9[b], N10-12[b], N13.6[b], N15.1, N15.9, N30-30.3[b], N30.8-30.9[b], N34-34.1[b], N39.0[b], N61, N70-76.8[b], N98.0, O07.0, O07.5, O08.0, O23-23.9, O41.1, O75.3, O85-86.8[b], O91-91.1, O98.0-98.2, P23.1-23.6, P36, P37.0, T80.2, T81.4, T82.6-82.7, T83.5-83.6, T84.5-84.7, T85.7, T88.0, Z22.0-22.3 |
| Viral | ICD-8 | 008.80-008.98, 040.00-043.99, 045.00-065.99, 067.00-072.09, 074.00-075.09, 078.00-079.20, 079.40-079.99, 099.92, 460.99, 464.01-480,99, 508.03, 761.20, 761.30 |
| | ICD-9 | 008H-M, 045-066, 070-072X, 074-075, 077-078H, 078W-079X, 279K, 321B-H, 323A, 323C-D, 460, 464-465X, 480-X, 487-W, 647F, G, 711F, 771A, B, 790W |
| | ICD-10 | A08-08.4, A60-60.9, A63.0, A80-89, A90-99, B00-06.0, B06.8-09, B15-19.9, B20-24, B25-34, B97-97.8, G02.0, G05.1, J00, J04-06.9[b], J10-11.8, J12-12.9, J20.3-20.7, J21.0, O35.3, O98.4-98.5, P23.0, P35, Z21, Z22.5-22.6 |

| | | |
|---|---|---|
| CNS | ICD-8 | 013.00-013.99, 027.01, 036.00, 090.40, 094.00-094.98, 320.00-320.80, 322.00-322.03, 392.99, 040.00-043.99, 045.00-046.99, 052.00, 054.04, 062.00-065.99, 071.99, 072.01, 075.02, 079.20, 474.99, 084.00, 320.88-320.99 |
| | ICD-9 | 013-X, 036A, B, 090E, 094-X, 320-X, 324-X, 392-X, 045-049X, 054D, 052B, 053A, 055A, 056A, 071, 072B, C, 321B-H, 323A, 323C, D, 006F, 321A, 321W |
| | ICD-10 | A02.2 (if G01), A17-17.9, A20.3, A22.8, A32.1, A39.0, A39.8 (if G05), A50.4 (if G05.0 or G01), A51.4 (if G01), A52.1 (if G05.0, G01 or F02.8), A54.8 (if G07 or G01), A69.2 (if G01), G00-00.9, G01, G04.2, G05.0, G06-06.2, G07, I02-02.9, A80-89, B00.3-00.4, B01.0-01.1, B02.0-02.1, B05.0-05.1, B06.0, B26.1-26.2, G02.0, G05.1, B58.2, A06.6, B37.5, B38.4, B43.1, B45.1, B46.1, B50.0, B57.4, B60.2, B69.0, B83.2, G02.1-02.8, G04, G04.9, G05.2 |
| Respiratory | ICD-8 | 010-012, 020.10, 461.00-461.09, 462.02, 463.01, 481,99-482.98, 501.99, 508.00-508.02, 510.01-510.09, 511.10, 513.99, 460.99, 464.01-464.09, 465.99, 470.99-473.99, 480,99, 508.03, 462.01, 462.09, 463.09, 466.99, 483.99-486.09, 502.00-503.09, 519.92, 490.99-491.09 |
| | ICD-9 | 010-012W, 031A, 033-034B, 052A, 055B, 112E, 122B, 460-466, 475, 481-482X, 510-X, 511B, 513-B, 480-X, 487-W, 462, 463, 466-B, 473-X, 483, 485, 486, 490, 491B |
| | ICD-10 | A15-16, A20.2, A21.2, A22.1, A31.0, A37, A38, A48.1, B00.2, B01.2, B05.2, B27, B37.1, B39-42, B44, B45.0, B46.0, B58.3, B59, J01-01.9, J02.0, J03.0, J13-15.9, J16.0, J20.0-20.2, J34.0, J36, J39.0-39.1, J85.1-85.3, J86-86.9, J00, J04-06.9, J10-11.8, J12-12.9, J20.3-20.7, J21.0, J02, J02.8-02.9, J03, J03.8-03.9, J16, J16.8, J18-18.9, J20, J20.8-21, J21.8-21.9, J22, J32-32.9, J35.0, J37-37.1, J40-42 |
| Skin | ICD-8 | 017.01-017.09, 110-111, 050-057, 680.00-680.90, 681.00-682.99, 684.00-684.09, 686.00-686.98 |
| | ICD-9 | 017A, 031B, 050-057, 074D, 091D, 110-111, 112D, 681-682X, 683, 684, 685-686X, 680A |
| | ICD-10 | A18.4, A20.0, A22.0, A26.0, A31.1, A32, A36.3, B00-09, B35-36, B37.2, B43.0, B43.2, B45.2, B46.3, B55.1, L00, L01-01.1, L02-02.9, L03-03.9, L04-08.9, L70.0, L30.3 |
| Genitourinary | ICD-8 | 090-099, 016, 054.02, 590.00-590.99, 595.00-595.02, 597.00, 599.02, 601.00, 604.00, 604.01, 607.30, 611.00, 611.01, 612.01-614.99, 616.00-616.03, 620.00-620.99, 622.00-622.19, 629.40 |
| | ICD-9 | 016, 054B, 112B, C, 090-099, 131A, 590-X, 597A, 595-D, X, 597W, 599A, 601-D, 603B, 604A, 604X, 607B, C, 608A, E, 611A, 614-F, W-X, 615A, X, 616-X |
| | ICD-10 | A18.0-18.1, A50-64, A70-74, B37.3-37.4, N10-12, N13.6, N15.1, N15.9, N30-30.3, N30.8-30.9, N34-34.1, N39.0, N41-41.3, N43.1, N45.0-45.9, N48.1-48.2, N49-49.9, N61, N70-76.8, N98.0 |
| Gastrointestinal | ICD-8 | 000-009, 014, 039.92, 127.99, 522.50, 527.30, 528.00, 528.30, 562.00-562.19, 566.00-566.01, 567.00-567.02, 569.00, 577.01, 540.00-540.99, 572,99 |
| | ICD-9 | 001-009, 123, 123, 127, 129, 014, 522E, H, 526E, 527D, 528A, D, 540B, 562-B, 566, 567-C, 569F, 575A, 540A, X, 572A |
| | ICD-10 | 001-009, 123, 123, 127, 129, 014, 522E, H, 526E, 527D, 528A, D, 540B, 562-B, 566, 567-C, 569F, 575A, 540A, X, 572A |

[a] In Sweden ICD-8 was used from 1969 to 1986, ICD-9 from 1987 to 1996, and ICD-10 from 1997 to date. All diagnoses of post-infection or sequel are excluded.

[b] The additional codes, B95-97.8, denotes the infecting organism. Bacteria= B95-96.8, Virus= B97- 97.8.

**Table S3.** Characteristics of the study sample, comparing those diagnosed with intellectual disability (ID) to those who were not.

| | | No ID diagnosis | Any ID diagnosis | p-value[a] |
|---|---|---|---|---|
| **N** | | 544,435 | 5,532 | |
| **Sex** | Female | 266,032 (48·9%) | 1,963 (35·5%) | <0·001 |
| | Male | 278,403 (51·1%) | 3,569 (64·5%) | |
| **Maternal psychiatric diagnosis** | Not present | 308,636 (56·7%) | 2,454 (44·4%) | <0·001 |
| | Present | 235,799 (43·3%) | 3,078 (55·6%) | |
| **Paternal psychiatric diagnosis** | Not present | 394,063 (72·4%) | 3,541 (64·0%) | <0·001 |
| | Present | 150,372 (27·6%) | 1,991 (36·0%) | |
| **Maternal age, years** | <25 | 80,114 (14·7%) | 1,155 (20·9%) | <0·001 |
| | 25-29 | 160,565 (29·5%) | 1,663 (30·1%) | |
| | 30-34 | 189,950 (34·9%) | 1,617 (29·2%) | |
| | 35-39 | 94,454 (17·3%) | 871 (15·7%) | |
| | >39 | 19,352 (3·6%) | 226 (4·1%) | |
| **Birth order** | First born | 245,164 (45·0%) | 2,256 (40·8%) | <0·001 |
| | Second born | 197,588 (36·3%) | 1,901 (34·4%) | |
| | Third born or later | 101,683 (18·7%) | 1,375 (24·9%) | |
| **Maternal BMI** | Underweight (<18·5) | 14,732 (2·7%) | 175 (3·2%) | <0·001 |
| | Normal weight (18·5 - < 25) | 276,488 (50·8%) | 2,293 (41·4%) | |
| | Overweight (25 – < 30) | 78,731 (14·5%) | 875 (15·8%) | |
| | Obese (>30) | 26,663 (4·9%) | 469 (8·5%) | |
| | Missing | 147,821 (27·2%) | 1,720 (31·1%) | |
| **Season of birth** | Dec-Feb | 127,698 (23·5%) | 1,329 (24·0%) | 0·110 |
| | Mar-May | 147,203 (27·0%) | 1,447 (26·2%) | |
| | Jun-Aug | 141,360 (26·0%) | 1,396 (25·2%) | |
| | Sep-Nov | 128,174 (23·5%) | 1,360 (24·6%) | |
| **Family income at birth** | Low (0-20%) | 76,514 (14·1%) | 1,352 (24·4%) | <0·001 |
| | Medium-low (20<-40%) | 114,293 (21·0%) | 1,462 (26·4%) | |
| | Medium (40<-60%) | 118,369 (21·7%) | 1,108 (20·0%) | |
| | Medium-high (60<-80%) | 117,903 (21·7%) | 850 (15·4%) | |
| | High (80<-100%) | 116,729 (21·4%) | 688 (12·4%) | |
| | Missing [b] | 627 (0·1%) | 72 (1·3%) | |
| **Highest parental education at birth** | <10 years | 32,172 (5·9%) | 718 (13·0%) | <0·001 |
| | 10-12 | 207,439 (38·1%) | 2,708 (49·0%) | |
| | >12 | 303,643 (55·8%) | 2,031 (36·7%) | |
| | Missing [b] | 1,181 (0·2%) | 75 (1·4%) | |
| **Maternal region of origin** | Sweden | 404,013 (74·2%) | 3,357 (60·7%) | <0·001 |
| | Nordics | 15,217 (2·8%) | 183 (3·3%) | |
| | West | 24,202 (4·4%) | 248 (4·5%) | |
| | Africa | 25,037 (4·6%) | 521 (9·4%) | |
| | Asia and Ocean | 11,481 (2·1%) | 165 (3·0%) | |

| | | | | |
|---|---|---|---|---|
| | Middle-East | 45,493 (8·4%) | 802 (14·5%) | |
| | East EU and Russia | 7,393 (1·4%) | 59 (1·1%) | |
| | South America | 11,518 (2·1%) | 197 (3·6%) | |
| | Unknown/Missing [b] | 81 (<1%) | 0 (0·0%) | |
| **Paternal region of origin** | Sweden | 410,409 (75·4%) | 3,532 (63·8%) | <0·001 |
| | Nordics | 19,360 (3·6%) | 201 (3·6%) | |
| | West | 15,896 (2·9%) | 167 (3·0%) | |
| | Africa | 20,863 (3·8%) | 455 (8·2%) | |
| | Asia and Ocean | 17,092 (3·1%) | 211 (3·8%) | |
| | Middle-East | 38,645 (7·1%) | 675 (12·2%) | |
| | East EU and Russia | 11,695 (2·1%) | 107 (1·9%) | |
| | South America | 10,458 (1·9%) | 184 (3·3%) | |
| | Unknown/Missing [b] | 17 (<1%) | 0 (0·0%) | |
| **Maternal infection during pregnancy** | No infection | 510,867 (93·8%) | 5,087 (92·0%) | <0·001 |
| | Pregnancy infection | 33,568 (6·2%) | 445 (8·0%) | |
| Maternal inpatient infection | No infection | 526,604 (96·7%) | 5,226 (94·5%) | <0·001 |
| | Pregnancy infection | 17,831 (3·3%) | 306 (5·5%) | |
| First trimester maternal infection | No infection | 538,337 (98·9%) | 5,463 (98·8%) | 0·370 |
| | Pregnancy infection | 6,098 (1·1%) | 69 (1·2%) | |
| Second trimester maternal infection | No infection | 535,126 (98·3%) | 5,403 (97·7%) | <0·001 |
| | Pregnancy infection | 9,309 (1·7%) | 129 (2·3%) | |
| Third trimester maternal infection | No infection | 525,438 (96·5%) | 5,265 (95·2%) | <0·001 |
| | Pregnancy infection | 18,997 (3·5%) | 267 (4·8%) | |
| Maternal bacterial infection | No infection | 515,753 (94·7%) | 5,127 (92·7%) | <0·001 |
| | Pregnancy infection | 28,682 (5·3%) | 405 (7·3%) | |
| Maternal viral infection | No infection | 532,273 (97·8%) | 5,369 (97·1%) | <0·001 |
| | Pregnancy infection | 12,162 (2·2%) | 163 (2·9%) | |
| Other/Unknown maternal infection | No infection | 525,151 (96·5%) | 5,287 (95·6%) | <0·001 |
| | Pregnancy infection | 19,284 (3·5%) | 245 (4·4%) | |
| **Paternal infection during pregnancy** | No infection | 536,342 (98·5%) | 5,440 (98·3%) | 0·280 |
| | Pregnancy infection | 8,093 (1·5%) | 92 (1·7%) | |
| **Maternal negative control infection** | No infection | 527,244 (96·8%) | 5,352 (96·7%) | 0·680 |
| | Pregnancy infection | 17,191 (3·2%) | 180 (3·3%) | |
| **Paternal negative control infection** | No infection | 536,720 (98·6%) | 5,455 (98·6%) | 0·870 |
| | Pregnancy infection | 7,715 (1·4%) | 77 (1·4%) | |

[a] p-values reported for $\chi^2$-test.
[b] Missing categories were recoded as Medium income, <10 years of education and born in Sweden.
Abbreviations: **BMI**: Body mass index; **ID**: Intellectual disability.

**Table S4.** Characteristics of the study sample, comparing those diagnosed with autism to those who were not.

| | | No autism diagnosis | Any autism diagnosis | p-value[a] |
|---|---|---|---|---|
| | | 535,809 | 14,158 | |
| **Sex** | Female | 263,828 (49·2%) | 4,167 (29·4%) | <0·001 |
| | Male | 271,981 (50·8%) | 9,991 (70·6%) | |
| **Maternal psychiatric diagnosis** | Not present | 305,768 (57·1%) | 5,322 (37·6%) | <0·001 |
| | Present | 230,041 (42·9%) | 8,836 (62·4%) | |
| **Paternal psychiatric diagnosis** | Not present | 388,973 (72·6%) | 8,631 (61·0%) | <0·001 |
| | Present | 146,836 (27·4%) | 5,527 (39·0%) | |
| **Maternal age, years** | <25 | 79,039 (14·8%) | 2,230 (15·8%) | <0·001 |
| | 25-29 | 158,151 (29·5%) | 4,077 (28·8%) | |
| | 30-34 | 186,917 (34·9%) | 4,650 (32·8%) | |
| | 35-39 | 92,747 (17·3%) | 2,578 (18·2%) | |
| | >39 | 18,955 (3·5%) | 623 (4·4%) | |
| **Birth order** | First born | 240,601 (44·9%) | 6,819 (48·2%) | <0·001 |
| | Second born | 194,783 (36·4%) | 4,706 (33·2%) | |
| | Third born or later | 100,425 (18·7%) | 2,633 (18·6%) | |
| **Maternal BMI** | Underweight (<18·5) | 14,545 (2·7%) | 362 (2·6%) | <0·001 |
| | Normal weight (18·5 - < 25) | 272,108 (50·8%) | 6,673 (47·1%) | |
| | Overweight (25 – < 30) | 77,075 (14·4%) | 2,531 (17·9%) | |
| | Obese (>30) | 25,959 (4·8%) | 1,173 (8·3%) | |
| | Missing | 146,122 (27·3%) | 3,419 (24·1%) | |
| **Season of birth** | Dec-Feb | 125,768 (23·5%) | 3,259 (23·0%) | <0·001 |
| | Mar-May | 145,077 (27·1%) | 3,573 (25·2%) | |
| | Jun-Aug | 139,048 (26·0%) | 3,708 (26·2%) | |
| | Sep-Nov | 125,916 (23·5%) | 3,618 (25·6%) | |
| **Family income at birth** | Low (0-20%) | 75,920 (14·2%) | 1,946 (13·7%) | <0·001 |
| | Medium-low (20<-40%) | 112,151 (20·9%) | 3,604 (25·5%) | |
| | Medium (40<-60%) | 116,105 (21·7%) | 3,372 (23·8%) | |
| | Medium-high (60<-80%) | 115,861 (21·6%) | 2,892 (20·4%) | |
| | High (80<-100%) | 115,089 (21·5%) | 2,328 (16·4%) | |
| | Missing [b] | 683 (0·1%) | 16 (0·1%) | |
| **Highest parental education at birth** | <10 years | 31,951 (6·0%) | 939 (6·6%) | <0·001 |
| | 10-12 | 204,113 (38·1%) | 6,034 (42·6%) | |
| | >12 | 298,509 (55·7%) | 7,165 (50·6%) | |
| | Missing[b] | 1,236 (0·2%) | 20 (0·1%) | |
| **Maternal region of origin** | Sweden | 396,654 (74·0%) | 10,716 (75·7%) | <0·001 |
| | Nordics | 14,958 (2·8%) | 442 (3·1%) | |
| | West | 23,909 (4·5%) | 541 (3·8%) | |
| | Africa | 24,929 (4·7%) | 629 (4·4%) | |
| | Asia and Ocean | 11,370 (2·1%) | 276 (1·9%) | |
| | Middle-East | 45,356 (8·5%) | 939 (6·6%) | |
| | East EU and Russia | 7,254 (1·4%) | 198 (1·4%) | |
| | South America | 11,302 (2·1%) | 413 (2·9%) | |

| | | | | |
|---|---|---|---|---|
| | Unknown/Missing[b] | 77 (<1%) | 4 (<1%) | |
| **Paternal region of origin** | Sweden | 402,980 (75·2%) | 10,961 (77·4%) | <0·001 |
| | Nordics | 19,084 (3·6%) | 477 (3·4%) | |
| | West | 15,708 (2·9%) | 355 (2·5%) | |
| | Africa | 20,826 (3·9%) | 492 (3·5%) | |
| | Asia and Ocean | 16,888 (3·2%) | 415 (2·9%) | |
| | Middle-East | 38,572 (7·2%) | 748 (5·3%) | |
| | East EU and Russia | 11,463 (2·1%) | 339 (2·4%) | |
| | South America | 10,271 (1·9%) | 371 (2·6%) | |
| | Unknown/Missing[b] | 17 (<1%) | 0 (0·0%) | |
| **Maternal infection during pregnancy** | No infection | 502,919 (93·9%) | 13,035 (92·1%) | <0·001 |
| | Pregnancy infection | 32,890 (6·1%) | 1,123 (7·9%) | |
| Maternal inpatient infection | No infection | 518,314 (96·7%) | 13,516 (95·5%) | <0·001 |
| | Pregnancy infection | 17,495 (3·3%) | 642 (4·5%) | |
| First trimester maternal infection | No infection | 529,826 (98·9%) | 13,974 (98·7%) | 0·041 |
| | Pregnancy infection | 5,983 (1·1%) | 184 (1·3%) | |
| Second trimester maternal infection | No infection | 526,689 (98·3%) | 13,840 (97·8%) | <0·001 |
| | Pregnancy infection | 9,120 (1·7%) | 318 (2·2%) | |
| Third trimester maternal infection | No infection | 517,197 (96·5%) | 13,506 (95·4%) | <0·001 |
| | Pregnancy infection | 18,612 (3·5%) | 652 (4·6%) | |
| Maternal bacterial infection | No infection | 507,720 (94·8%) | 13,160 (93·0%) | <0·001 |
| | Pregnancy infection | 28,089 (5·2%) | 998 (7·0%) | |
| Maternal viral infection | No infection | 523,945 (97·8%) | 13,697 (96·7%) | <0·001 |
| | Pregnancy infection | 11,864 (2·2%) | 461 (3·3%) | |
| Other/Unknown maternal infection | No infection | 516,954 (96·5%) | 13,484 (95·2%) | <0·001 |
| | Pregnancy infection | 18,855 (3·5%) | 674 (4·8%) | |
| **Paternal infection during pregnancy** | No infection | 527,869 (98·5%) | 13,913 (98·3%) | 0·016 |
| | Pregnancy infection | 7,940 (1·5%) | 245 (1·7%) | |
| **Maternal negative control infection** | No infection | 519,005 (96·9%) | 13,591 (96·0%) | <0·001 |
| | Pregnancy infection | 16,804 (3·1%) | 567 (4·0%) | |
| **Paternal negative control infection** | No infection | 528,214 (98·6%) | 13,961 (98·6%) | 0·800 |
| | Pregnancy infection | 7,595 (1·4%) | 197 (1·4%) | |

[a] p-values reported for $\chi^2$-test.
[b] Missing categories were recoded as Medium income, <10 years of education and born in Sweden.
Abbreviations: **BMI**: Body mass index.

**Table S5.** Characteristics of the sibling sample for the analysis of intellectual disability (ID) as an outcome, comparing individuals with any ID diagnosis to their siblings without ID, and sibling pairs where none of the siblings were diagnosed with autism.

| | | **Contribute to the Matched Sibling Analysis** | | **Does not Contribute to the Matched Sibling Analysis** |
|---|---|---|---|---|
| | | **No ID diagnosis** | **Any ID diagnosis** | **Undiagnosed pairs** |
| **N** | | 5,338 | 3,937 | 384,818 |
| **Sex** | Female | 2,617 (49·0%) | 1,366 (34·7%) | 187,722 (48·8%) |
| | Male | 2,721 (51·0%) | 2,571 (65·3%) | 197,096 (51·2%) |
| **Maternal psychiatric diagnosis** | Not present | 2,627 (49·2%) | 1,825 (46·4%) | 223,985 (58·2%) |
| | Present | 2,711 (50·8%) | 2,112 (53·6%) | 160,833 (41·8%) |
| **Paternal psychiatric diagnosis** | Not present | 3,577 (67·0%) | 2,569 (65·3%) | 283,115 (73·6%) |
| | Present | 1,761 (33·0%) | 1,368 (34·7%) | 101,703 (26·4%) |
| **Maternal age, years** | <25 | 1,052 (19·7%) | 784 (19·9%) | 53,488 (13·9%) |
| | 25-29 | 1,656 (31·0%) | 1,269 (32·2%) | 118,672 (30·8%) |
| | 30-34 | 1,621 (30·4%) | 1,182 (30·0%) | 139,195 (36·2%) |
| | 35-39 | 829 (15·5%) | 585 (14·9%) | 63,290 (16·4%) |
| | >39 | 180 (3·4%) | 117 (3·0%) | 10,173 (2·6%) |
| **Birth order** | First born | 1,594 (29·9%) | 1,341 (34·1%) | 148,097 (38·5%) |
| | Second born | 1,807 (33·9%) | 1,556 (39·5%) | 161,456 (42·0%) |
| | Third born or later | 1,937 (36·3%) | 1,040 (26·4%) | 75,265 (19·6%) |
| **Maternal BMI** | Underweight | 159 (3·0%) | 119 (3·0%) | 9,796 (2·5%) |
| | Normal weight | 2,132 (39·9%) | 1,648 (41·9%) | 196,313 (51·0%) |
| | Overweight | 978 (18·3%) | 638 (16·2%) | 55,977 (14·5%) |
| | Obese | 533 (10·0%) | 339 (8·6%) | 18,495 (4·8%) |
| | Missing | 1,536 (28·8%) | 1,193 (30·3%) | 104,237 (27·1%) |
| **Season of birth** | Dec-Feb | 1,326 (24·8%) | 917 (23·3%) | 89,839 (23·3%) |
| | Mar-May | 1,389 (26·0%) | 1,042 (26·5%) | 105,469 (27·4%) |
| | Jun-Aug | 1,381 (25·9%) | 1,015 (25·8%) | 99,899 (26·0%) |
| | Sep-Nov | 1,242 (23·3%) | 963 (24·5%) | 89,611 (23·3%) |
| **Maternal infection during pregnancy** | No infection | 4,973 (93·2%) | 3,656 (92·9%) | 363,148 (94·4%) |
| | Pregnancy infection | 365 (6·8%) | 281 (7·1%) | 21,670 (5·6%) |
| **Maternal inpatient infection** | No infection | 5,121 (95·9%) | 3,749 (95·2%) | 373,389 (97·0%) |
| | Pregnancy infection | 217 (4·1%) | 188 (4·8%) | 11,429 (3·0%) |
| **Paternal infection during pregnancy** | No infection | 5,239 (98·1%) | 3,869 (98·3%) | 379,257 (98·6%) |
| | Pregnancy infection | 99 (1·9%) | 68 (1·7%) | 5,561 (1·4%) |
| **Paternal inpatient infection** | No infection | 5,299 (99·3%) | 3,899 (99·0%) | 382,864 (99·5%) |
| | Pregnancy infection | 39 (0·7%) | 38 (1·0%) | 1,954 (0·5%) |

Abbreviations: **BMI**: Body mass index; **ID**: Intellectual disability.

**Table S6.** Characteristics of the sibling sample for the analysis of autism as an outcome, comparing individuals with any autism diagnosis to their siblings without autism, and sibling pairs where none of the siblings were diagnosed with autism.

| | | **Contribute to the Matched Sibling Analysis** | | **Does not contribute to the Matched Sibling Analysis** |
|---|---|---|---|---|
| | | **No autism diagnosis** | **Any autism diagnosis** | |
| **N** | | 11,834 | 10,030 | 372,229 |
| **Sex** | Female | 6,018 (50·9%) | 2,926 (29·2%) | 182,761 (49·1%) |
| | Male | 5,816 (49·1%) | 7,104 (70·8%) | 189,468 (50·9%) |
| **Maternal psychiatric diagnosis** | Not present | 5,059 (42·7%) | 3,981 (39·7%) | 219,397 (58·9%) |
| | Present | 6,775 (57·3%) | 6,049 (60·3%) | 152,832 (41·1%) |
| **Paternal psychiatric diagnosis** | Not present | 7,639 (64·6%) | 6,266 (62·5%) | 275,356 (74·0%) |
| | Present | 4,195 (35·4%) | 3,764 (37·5%) | 96,873 (26·0%) |
| **Maternal age, years** | <25 | 1,882 (15·9%) | 1,422 (14·2%) | 52,020 (14·0%) |
| | 25-29 | 3,417 (28·9%) | 3,062 (30·5%) | 115,118 (30·9%) |
| | 30-34 | 3,964 (33·5%) | 3,536 (35·3%) | 134,498 (36·1%) |
| | 35-39 | 2,156 (18·2%) | 1,695 (16·9%) | 60,853 (16·3%) |
| | >39 | 415 (3·5%) | 315 (3·1%) | 9,740 (2·6%) |
| **Birth order** | First born | 3,810 (32·2%) | 4,097 (40·8%) | 143,125 (38·5%) |
| | Second born | 4,728 (40·0%) | 3,929 (39·2%) | 156,162 (42·0%) |
| | Third born or later | 3,296 (27·9%) | 2,004 (20·0%) | 72,942 (19·6%) |
| **Maternal BMI** | Underweight | 300 (2·5%) | 239 (2·4%) | 9,535 (2·6%) |
| | Normal weight | 5,548 (46·9%) | 4,728 (47·1%) | 189,817 (51·0%) |
| | Overweight | 2,049 (17·3%) | 1,822 (18·2%) | 53,722 (14·4%) |
| | Obese | 965 (8·2%) | 836 (8·3%) | 17,566 (4·7%) |
| | Missing | 2,972 (25·1%) | 2,405 (24·0%) | 101,589 (27·3%) |
| **Season of birth** | Dec-Feb | 2,779 (23·5%) | 2,313 (23·1%) | 86,990 (23·4%) |
| | Mar-May | 3,202 (27·1%) | 2,586 (25·8%) | 102,112 (27·4%) |
| | Jun-Aug | 3,150 (26·6%) | 2,621 (26·1%) | 96,524 (25·9%) |
| | Sep-Nov | 2,703 (22·8%) | 2,510 (25·0%) | 86,603 (23·3%) |
| **Maternal infection during pregnancy** | No infection | 11,037 (93·3%) | 9,337 (93·1%) | 351,403 (94·4%) |
| | Pregnancy infection | 797 (6·7%) | 693 (6·9%) | 20,826 (5·6%) |
| **Maternal inpatient infection** | No infection | 11,439 (96·7%) | 9,629 (96·0%) | 361,191 (97·0%) |
| | Pregnancy infection | 395 (3·3%) | 401 (4·0%) | 11,038 (3·0%) |
| **Paternal infection during pregnancy** | No infection | 11,618 (98·2%) | 9,858 (98·3%) | 366,889 (98·6%) |
| | Pregnancy infection | 216 (1·8%) | 172 (1·7%) | 5,340 (1·4%) |
| **Paternal inpatient infection** | No infection | 11,756 (99·3%) | 9,972 (99·4%) | 370,334 (99·5%) |
| | Pregnancy infection | 78 (0·7%) | 58 (0·6%) | 1,895 (0·5%) |

Abbreviations: **BMI**: Body mass index.

**Table S7**. Summary of model estimates (hazard ratios) using Cox regression for risk of intellectual disability (ID) or autism, by categories of specific maternal infections.

| | Maternal Infections | | | | |
|---|---|---|---|---|---|
| | Events (Cases) | Person-years | Model 1[a] | Model 2[b] | Model 3[c] |
| ID with bacterial infection | | | | | |
| No Pregnancy Infection | 5,127 | 7,077,679 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 405 | 329,203 | 1·74 (1·57; 1·92) | 1·42 (1·28; 1·58) | 1·42 (1·28; 1·58) |
| Autism with bacterial infection | | | | | |
| No Pregnancy Infection | 13,159 | 7,049,467 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 998 | 327,243 | 1·36 (1·27; 1·45) | 1·19 (1·12; 1·27) | 1·18 (1·10; 1·26) |
| ID with viral infection | | | | | |
| No Pregnancy Infection | 5,369 | 7,269,009 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 163 | 137,873 | 1·63 (1·39; 1·90) | 1·30 (1·11; 1·53) | 1·30 (1·10; 1·52) |
| Autism with viral infection | | | | | |
| No Pregnancy Infection | 13,696 | 7,239,684 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 461 | 137,025 | 1·48 (1·34; 1·62) | 1·25 (1·14; 1·38) | 1·23 (1·12; 1·35) |
| ID with other/unknown infection | | | | | |
| No Pregnancy Infection | 5,287 | 7,191,273 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 245 | 215,608 | 1·58 (1·39; 1·80) | 1·32 (1·16; 1·50) | 1·31 (1·15; 1·50) |
| Autism with other/unknown infection | | | | | |
| No Pregnancy Infection | 13,483 | 7,162,490 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 674 | 214,220 | 1·37 (1·26; 1·48) | 1·20 (1·11; 1·30) | 1·18 (1·09; 1·28) |

[a] Adjusted for sex and birth year.
[b] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education and income.
[c] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education, income and any infection during pre-pregnancy.
Abbreviations: **Ref.**: Reference; **ID**: Intellectual disability.

**Table S8**. Summary of model estimates (hazard ratios) using Cox regression for risk of intellectual disability (ID) or autism, by categories of maternal infection status for different trimesters.

| | Maternal Infections | | | | |
|---|---|---|---|---|---|
| | Events (Cases) | Person-years | Model 1[a] | Model 2[b] | Model 3[c] |
| ID (1st trimester infection) | | | | | |
| No Pregnancy Infection | 5,463 | 7,343,036 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 69 | 63,846 | 1·51 (1·19; 1·92) | 1·14 (0·90; 1·45) | 1·13 (0·89; 1·44) |
| Autism (1st trimester infection) | | | | | |
| No Pregnancy Infection | 13,973 | 7,313,197 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 184 | 63,513 | 1·20 (1·04; 1·39) | 1·06 (0·91; 1·22) | 1·03 (0·89; 1·20) |
| ID (2nd trimester infection) | | | | | |
| No Pregnancy Infection | 5,403 | 7,307,004 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 129 | 99,878 | 1·81 (1·51; 2·15) | 1·45 (1·22; 1·74) | 1·45 (1·21; 1·74) |
| Autism (2nd trimester infection) | | | | | |
| No Pregnancy Infection | 13,839 | 7,277,435 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 318 | 99,275 | 1·36 (1·22; 1·52) | 1·21 (1·08; 1·35) | 1·19 (1·06; 1·33) |
| ID (3rd trimester infection) | | | | | |
| No Pregnancy Infection | 5,265 | 7,183,888 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 267 | 222,994 | 1·66 (1·46; 1·87) | 1·41 (1·25; 1·60) | 1·41 (1·24; 1·60) |
| Autism (3rd trimester infection) | | | | | |
| No Pregnancy Infection | 13,505 | 7,155,030 | Ref. | Ref. | Ref. |
| Pregnancy Infection | 652 | 221,679 | 1·32 (1·22; 1·43) | 1·19 (1·10; 1·28) | 1·17 (1·08; 1·27) |

[a] Adjusted for sex and birth year.
[b] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education and income.
[c] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education, income and any infection during pre-pregnancy.
Abbreviations: **Ref.**: Reference; **ID**: Intellectual disability.

**Table S9**. Summary of model estimates (hazard ratios) using Cox regression for risk of intellectual disability (ID) or autism, by using only complete cases.

| | Maternal Infections | | | | | |
|---|---|---|---|---|---|---|
| | Events (Cases) | Person-years | Model 1[a] | Model 2[b] | Model 3[c] | Sibling Model[d] |
| ID | | | | | | |
| No Pre-pregnancy infection | 3,635 | 4,995,235 | Ref. | Ref. | | |
| Pre-pregnancy infection | 134 | 144,475 | 1·31 (1·10; 1·56) | 1·07 (0·90; 1·28) | | |
| | | | | | | |
| No Pregnancy Infection | 3,441 | 4,846,201 | Ref. | Ref. | Ref. | Ref. |
| Pregnancy Infection | 328 | 293,509 | 1·61 (1·43; 1·80) | 1·33 (1·19; 1·49) | 1·33 (1·18; 1·49) | 1·08 (0·83; 1·40) |
| ASD | | | | | | |
| No Pre-pregnancy infection | 10,268 | 4,971,496 | Ref. | Ref. | | |
| Pre-pregnancy infection | 454 | 143,460 | 1·37 (1·24; 1·50) | 1·24 (1·13; 1·36) | | |
| | | | | | | |
| No Pregnancy Infection | 9,832 | 4,823,263 | Ref. | Ref. | Ref. | Ref. |
| Pregnancy Infection | 890 | 291,694 | 1·31 (1·22; 1·40) | 1·15 (1·08; 1·24) | 1.14 (1.07; 1.23) | 0·92 (0·77; 1·08) |

[a] Adjusted for sex and birth year.
[b] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education and income.
[c] Adjusted for sex, birth order, birth year, BMI, maternal age at birth, season of birth, maternal and paternal regions of origin and psychiatric history, paternal infections during the same time period, parental education, income and any infection during pre-pregnancy.
[d] Sibling models adjusted for paternal infection, sex, birth order, year of birth (continuous) and season of birth.
Abbreviations: **Ref.**: Reference; **ID**: Intellectual disability.

**Table S10.** Evaluation of the potential carry-over effects for the sibling analysis.

| | Cases | HR (95% CI) |
|---|---|---|
| Any ID diagnosis | | |
| Older child not exposed | 68 | Ref. |
| Younger child exposed | 68 | 1·28 (0·88; 1·88) |
| | | |
| Older child exposed | 87 | 1·27 (0·90; 1·80) |
| Younger child not exposed | 64 | Ref. |
| Any autism diagnosis | | |
| Older child not exposed | 237 | Ref. |
| Younger child exposed | 173 | 1·03 (0·82; 1·30) |
| | | |
| Older child exposed | 223 | 0·94 (0·75; 1·19) |
| Younger child not exposed | 168 | Ref. |

Sibling models adjusted for paternal infection, sex and season of birth.
Abbreviations: **Ref.**: Reference; **ID**: Intellectual disability